

# High or low: a balanced benchmark for the seasonality of YouTube search across the advertising taxonomy

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July 19, 2026

**Abstract** In plain words: we watch how often the world searches YouTube for each of three thousand everyday topics, month by month for seventeen years, and ask a simple question about the two years we kept hidden — will this topic’s coming month be a *high* month or a *low* month? The question is built so that guessing scores exactly half. Simple memory — assume next month looks like the last one you saw — already answers correctly 92% of the time; the best possible score any method of this kind could reach is 96%; and the honest finding is that *nothing we built beats that simple memory*: the best learned models tie it within noise, and the remaining distance to the ceiling is quantisation noise at the bar, not hidden seasonality. Seasonal pattern machinery matters only for the minority of topics whose seasons genuinely cross the bar — and even there, only literal seasonal copying survives.

In one sentence: when the direction question is rebalanced — not *will this topic rise or fall?* but *will it sit high or low against one global bar?* — the predictability of worldwide YouTube-search interest jumps from  $\approx 0.59$  to  $0.9182$ , the measured ceiling is  $0.9635$ , and almost everything in between is memory, not seasonality.

We take the 3,021 audit-clean topics extracted from Google’s own advertising taxonomy, each with 210 months of worldwide YouTube-search interest (2008-01–2025-06), and define one balanced binary task. The label of a topic-month is  $+1$  when its monthly rate exceeds a single global threshold  $\tau = 19$  — the median rate over all topics  $\times$  training months only — and  $-1$  otherwise. By construction the classes split  $49.42\%/50.58\%$  on the training years (the closest an integer-valued rate allows), so a coin scores  $0.5$  exactly, and no class-imbalance artefact can flatter any model. The previous rise/fall formulation cannot be balanced at all: its tie mass at zero change forces a  $\approx 37/63$  split, which motivated this redefinition.

The protocol is walls-first — a *wall* is a cut in time that models may never look across: a training era, a 24-month validation window used only for selection, and a 24-month test window touched once. Each competing predictor is an *arm*; accuracy is reported per *horizon* (how many months past the wall the prediction reaches) and averaged over the 24 horizons. Two oracle bounds — scores available only to an all-knowing reference, hence unreachable ceilings — are measured before any model runs: a per-(topic, month) majority oracle caps all in-protocol prediction at  $0.9635$ , and cross-year label agreement — what perfect year-copying would score — is  $0.9270$ . The honest ladder then reads: carrying the last pre-wall state (persistence)  $0.9194$ ; the tiled seasonal naive  $0.9106$ ; the topic’s base rate  $0.7165$  and monthly climatology  $0.7145$ ; a phase encoder–decoder — the architecture at the centre of our previous rise/fall study —  $0.8568$  pure and  $0.8956$  with memory features; a direct multi-horizon pooled logistic  $0.9193$ ; and the champion, a per-topic selected ensemble over five arms,  $0.9183$  (normalised AUC over the 24 horizons; plain monthly mean  $0.9182$ ), with the selector choosing persistence for 2,767 topics, seasonal copy for 114, a three-year vote for 106, the direct model for 23 and the encoder–decoder for 11. We report the selection landscape honestly: selector families spanning two to five arms land within  $0.918$ – $0.922$  on test and their test ranking *inverts* their validation ranking — the differences are noise, and we ship the validation-chosen variant, not the test-best one. The lesson of the balanced lens is structural: on a level-persistent target the celebrated seasonal machinery survives only inside a selector, for the 11 topics that genuinely need it, while the gap that remains ( $0.9182 \rightarrow 0.9635$ ) lives in topics that flicker within one or two units of  $\tau$  — noise no interior model can call, and the natural frontier for exogenous signal. Every number regenerates from the registered open dataset and bundle; the champion’s 24-month

run is live as the reference entry of a fully public platform challenge whose leaderboard score is exactly this paper's 0.9182.

**Keywords** — seasonality; YouTube search; Google Trends; balanced classification; time walls; forecasting baselines; model selection; reproducibility

## 1 From rise/fall to high/low: why balance forces a new task

Our previous study asked, for every topic in the advertising taxonomy, whether next month's worldwide YouTube-search interest would rise or fall, and found a hard, honest answer: a family of pooled predictors reaches  $\approx 0.59$ – $0.60$  horizon-averaged accuracy where weighted memory reaches 0.5801, against a measured cap of 0.8322. That task has a structural flaw no model can fix: monthly Trends rates are small integers, so *no change* is a fat atom of the label distribution. Excluding ties (as we did) still leaves the realised classes near 37/63 across the population, and any accuracy number silently rides that imbalance. The balanced reformulation asks a cleaner question. Fix one global bar  $\tau$  and ask, for every topic and month: *does this topic's rate sit above the bar?* With  $y_{i,t}$  the worldwide monthly rate of topic  $i$  in month  $t$ ,

$$L_{i,t} = \begin{cases} +1 & y_{i,t} > \tau \\ -1 & \text{otherwise,} \end{cases} \quad \tau = \text{median}\{y_{i,t} : t \in \text{train}\}, \quad (1)$$

computed on the training era only, so the held-out years never leak into the label definition. This splits the dataset 50/50 by construction. The realised training split is 49.42% high (the integer atom at  $\tau = 19$  blocks exactness by half a percent; the adjacent threshold lands 50.85% on the other side). A coin scores 0.5; every point above it is earned.

The consequence of the redefinition is dramatic and diagnostic. Interest *levels* are strongly persistent where interest *changes* are nearly martingale: the same series that supported only 0.59 direction accuracy support 0.9182 high/low accuracy over the same held-out months — a genuinely different, and structurally more revealing, question about seasonality: which topics live above the world's median attention, when, and for how long.

## 2 Dataset, walls and ceilings

The vocabulary is the complete advertising content taxonomy Google publishes [3] (accessed 2026-07-13; the exact snapshot ships verbatim in the dataset as `verticals.csv`, which is the citable provenance), split into lowercase word-topics; after the audit gates (native monthly coverage, positivity, the published blacklist) 3,021 topics remain, each a 210-month worldwide monthly YouTube-search series — search interest as an observable of collective attention in the Google-Trends tradition [1] — over 2008-01–2025-06 (the most recent year of raw data is excluded from the study entirely, against recency artefacts). The walls are unchanged from the previous study: everything before the last 48 months trains; months  $-48..-25$  are validation, used only for selection and early stopping; the final 24 months are test, touched once. All 72,504 test cells are scored — the balanced label has no ties to exclude. The balance is defined on the training years only, and the realised *test* window is 53.13% high — levels drifted upward over the held-out years. This cannot flatter any headline: a random  $\pm 1$  coin still scores 0.5 regardless of imbalance, and fully exploiting the base rate reaches only 0.7165 — the champion's 0.9182 is driven by level-persistence (persistence alone scores 0.9194), not by test-set composition.

Two bounds are measured *before* any model runs. First, the per-(topic, month) majority oracle: knowing, for every topic and calendar month, which class that cell takes most often across the two test years caps every in-protocol predictor at 0.9635. Second, cross-year agreement: the fraction of test cells whose label equals the same topic-month one year later is 0.9270 — the score of perfect year-copying, and a

| arm                                   | ACC (AUC over horizons) | kind      |
|---------------------------------------|-------------------------|-----------|
| per-(topic,month) oracle              | 0.9635                  | ceiling   |
| cross-year agreement                  | 0.9270                  | ceiling   |
| <b>champion: per-topic selected-5</b> | <b>0.9183</b>           | selector  |
| direct pooled logistic                | 0.9193                  | learned   |
| persistence                           | 0.9194                  | reference |
| seasonal naive (tiled)                | 0.9106                  | reference |
| encoder–decoder + memory              | 0.8956                  | learned   |
| pooled logistic (AR rollout)          | 0.8908                  | learned   |
| encoder–decoder (pure phase)          | 0.8568                  | learned   |
| memory majority                       | 0.7286                  | reference |
| base rate                             | 0.7165                  | reference |
| label climatology                     | 0.7145                  | reference |
| coin                                  | 0.5000                  | floor     |

**TABLE 1.** The balanced scoreboard: horizon-averaged accuracy (normalised AUC over the 24 held-out months, coin = 0.5) on all 72,504 test cells. Machine-regenerated by `reproduce.py` from the registered dataset (CPU, seed 7, 4-decimal contract).

repeatability measure of the label itself. The champion below sits within 0.9 points of the second bound; the distance to the first is dominated by topics that flicker within one or two units of  $\tau$ , where the label is effectively noise to any model that sees only the series.

### 3 The ladder: memory first, machinery second

Table 1 is the complete scoreboard; every row regenerates from the registered dataset via the bundle contract. The headline metric is the trapezoid-normalised area under the accuracy–horizon curve,

$$\text{ACC} = \frac{1}{H-1} \int_1^H \text{acc}(h) dh, \quad H = 24, \quad (2)$$

with  $\text{acc}(h)$  the fraction of topics called correctly  $h$  months past the wall — the trapezoid weighting is why the AUC (0.9183) and the plain monthly mean (0.9182) differ in the last digit. The oracle ceiling is the per-(topic, month) majority bound

$$\text{O}_3 = \frac{1}{24N} \sum_{i,m} \max(n_{i,m}^+, n_{i,m}^-), \quad (3)$$

where  $n_{i,m}^\pm$  count the two test years’ labels of topic  $i$  in calendar month  $m$  — what a clairvoyant per-cell majority could score, 0.9635 here. The references are strong by construction on a persistent target: carrying the last pre-wall label forward scores 0.9194; the tiled seasonal naive [4] (copy the last observed pre-wall year) 0.9106; the topic’s base rate 0.7165; monthly climatology 0.7145.

Three learned arms compete above the references. The **phase encoder–decoder** — the previous study’s central architecture, retargeted at the high/low wave — attends [7] over the 36-month pre-wall wave, estimates each topic’s circular seasonal phase (anchored to the calendar), rotates, and classifies each queried month; it scores 0.8568 pure and 0.8956 with strictly pre-wall memory features. On this target the celebrated machinery *loses to raw persistence*: a result we report with the same prominence we gave its win on the rise/fall task. The **direct pooled logistic** predicts each of the 24 horizons straight from pre-wall features (wall state, tiled seasonal lags, three-year vote, level distance from  $\tau$ , base rate, horizon interactions, month one-hots) with no rollout and therefore no error compounding — the direct multi-horizon strategy of the forecasting literature [2, 5]; it reaches 0.9193, edging persistence’s curve at short horizons.

| arm             | topics | champion there | persistence there | $\Delta$ |
|-----------------|--------|----------------|-------------------|----------|
| persistence     | 2,767  | 0.9350         | 0.9350            | —        |
| seasonal copy   | 114    | 0.7566         | 0.6959            | +6.1pp   |
| three-year vote | 106    | 0.7178         | 0.8113            | −9.4pp   |
| direct          | 23     | 0.6848         | 0.7047            | −2.0pp   |
| encoder–decoder | 11     | 0.7879         | 0.8333            | −4.5pp   |
| all swapped     | 254    | 0.7352         | 0.7508            | −1.6pp   |

**TABLE 2.** Per-arm generalisation on test: mean accuracy over the topics each arm was chosen for, versus simply carrying persistence there. Only the seasonal copy earns its swap; machine-regenerated by `reproduce.py` on every run.

The **champion** — the validation winner we ship, which on test ties the strongest references within noise (Table 1) — is the simplest structure in the family: per-topic selection. Five candidate arms — persistence, seasonal copy, a three-year monthly vote, the direct model, the encoder–decoder — compete on the 24 validation months; each topic is then called on test by its validation winner. The selector chooses persistence for 2,767 of 3,021 topics, seasonal copy for 114, the vote for 106, the direct model for 23, and the encoder–decoder (the memory variant) for 11; the ensemble scores 0.9183 (plain monthly mean 0.9182). Fig. 1 shows the accuracy–horizon curves; Fig. 2 the arm shares.

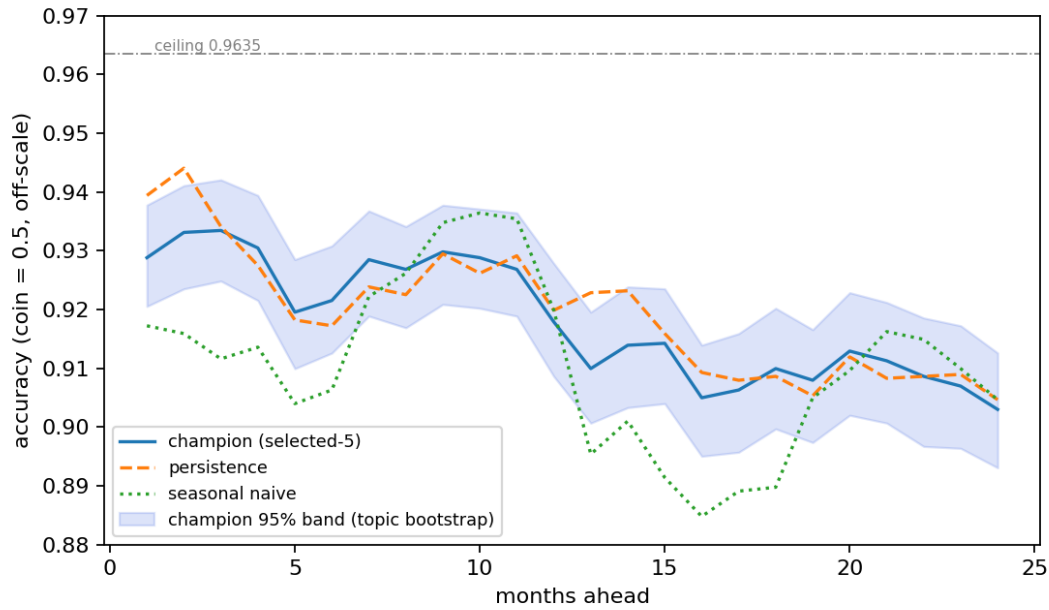
**Selection honesty.** Selector families are themselves a hyper-choice, and we report the landscape rather than the best cell of it: selectors spanning two to five arms, fit on half or all of validation, land within 0.918–0.922 on test, and their *test* ranking inverts their *validation* ranking — a half-validation comparison preferred the five-arm selector (0.9112 vs 0.9110 vs 0.9075 on the held-back validation half) while test mildly preferred fewer arms. These differences are noise, and now quantified: a topic-level bootstrap (resampling the 3,021 topics,  $B = 2000$ ) puts the champion at 0.9183 [0.9118, 0.9248] and persistence at 0.9194 [0.9124, 0.9262] — overlapping intervals that contain every selector variant; Fig. 1 overlays the champion’s band. We ship the validation-chosen five-arm variant and state plainly that nothing in this family is distinguishable from its siblings. The ladder is also robust to the only other defensible threshold: at  $\tau > 18$  (the integer atom’s other side, 50.85% high on train) the board barely moves — persistence 0.9194, seasonal naive 0.9109, direct 0.9208, selected ensemble 0.9180 — regenerable via `reproduce.py --tau-sens`.

#### 4 What the balanced lens shows

**Persistence dominates; only seasonal copying survives the wall.** 0.9194 of the achievable 0.9635 is reached by carrying one bit per topic across the wall, and no learned arm beats it on test — the recurring M-competition pattern of strong simple baselines [5], which persists even where machine learning takes the podium [6]. Table 2 shows what the selector’s swaps are actually worth: on the 254 topics where it swaps away from persistence, the champion scores 0.7352 against persistence’s 0.7508 there — a net loss. The gain concentrates entirely in the *seasonal-copy* arm (+6.1pp on its 114 topics); the vote (−9.4pp), the direct model (−2.0pp) and the phase encoder–decoder (−4.5pp) do not survive the wall. Even inside the seasonal residual, only literal seasonal copying carries seasonal signal across time; the selector’s other choices are validation noise. The per-arm table is machine-regenerated by the contract on every run.

**The remaining gap is a noise floor, not a modelling failure (Fig. 3).** The distance from 0.9182 to 0.9635 concentrates in topics whose rates hover within one or two integer units of  $\tau$ : their labels flip on quantisation jitter that no function of the series’ own past can call. Closing that gap requires exogenous signal — the same frontier our rise/fall study identified at its own (much lower) ceiling.

**One number, everywhere.** The champion’s 24-month autoregressed run is simultaneously this paper’s headline, the per-topic prediction chart on the host platform’s live topics pages, and the reference entry of a fully public platform challenge (24 future months, average monthly accuracy, no hidden holdout — the



**FIGURE 1.** Accuracy by horizon over the 24 held-out months: the champion (with its topic-bootstrap 95% band), its two strongest references, and the measured per-(topic,month) ceiling (0.9635, annotated). The y-axis is zoomed to [0.88, 0.97] so the arms separate; the coin floor (0.5) is far off-scale below. Machine-generated by `reproduce.py --fig1`.

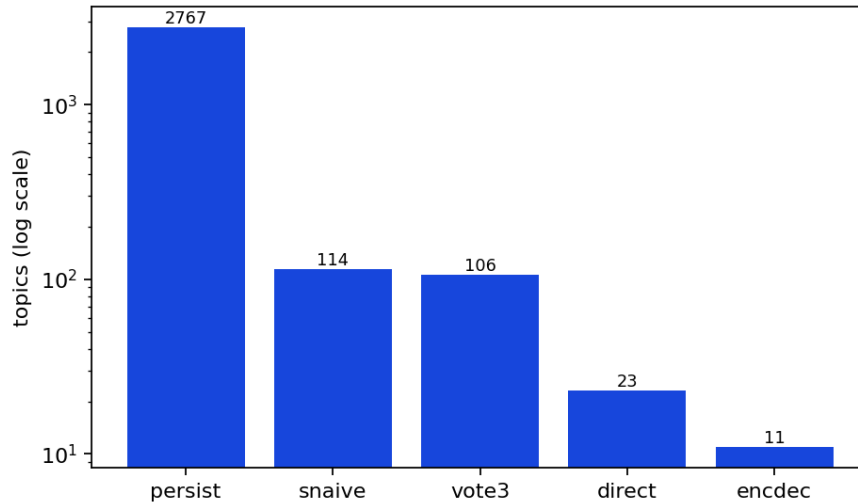
public leaderboard decides): all three surfaces display exactly 0.9182, regenerated from the same registered artifacts. The platform’s shipped predictions are included in the dataset (`platform_predictions.json`) and are bit-identical to the CPU contract run’s — `reproduce.py` prints the cell-by-cell comparison on every run (0 of 72,504 cells differ).

## 5 Reproduction contract

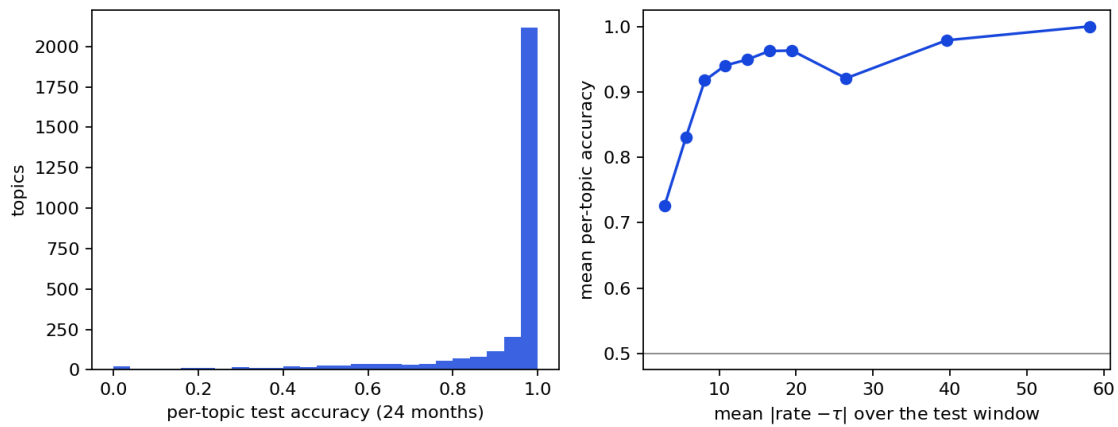
`reproduce.py <adsttopics-dataset-balanced.zip>` rebuilds the bench from the zip alone — the clean window, the audit gates,  $\tau$  from training cells only — and regenerates the ceilings, the references, the three learned arms and the champion, comparing every number to the registered `expected.json` at four decimals (CPU-canonical, seed 7, deterministic). `--fig1/--fig2/--fig3` regenerate the manuscript’s three figures; `--selftest` replaces the labels with a planted per-topic seasonal rule and verifies the protocol recovers it exactly. The dataset zip itself carries the labels, the threshold, the atlas, the full scoreboard and the complete research source (`src/balanced.py`), so the study is readable as well as re-runnable.

## 6 Relation to the rise/fall study

The previous submission asked the change-direction question and answered it honestly:  $\approx 0.59$  against a 0.8322 cap, with the phase encoder–decoder as champion. This study neither supersedes that result nor inflates it — it changes the question to one that *can* be balanced, and reports that the same series are substantially more predictable through the new lens — the error rate falls five-fold, from 41% on rise/fall to 8.2% on high/low — for reasons that are structural (level persistence), not architectural. The two scoreboards together bound what interior models can know about this vocabulary: changes are nearly martingale; states are nearly deterministic; seasonality lives in the crossings.



**FIGURE 2.** The champion’s per-topic arm choices (validation-fit selector), log scale so the four small arms are readable beside persistence’s 2,767: seasonal machinery serves the strongly seasonal residual. Machine-generated by `reproduce.py --fig2`.



**FIGURE 3.** The noise floor made visible. Left: the distribution of per-topic test accuracy — most topics are called almost perfectly; the thin tail includes topics scored near zero, i.e. persistently *miscalled* (a wrong-side wall state carried 24 months), not merely hard. Right: decile-mean per-topic accuracy against the topic’s mean distance  $|\text{rate} - \tau|$  over the test window — the decile means fall to  $\approx 0.73$  where series hover within one or two integer units of the bar (individual topics there reach the coin and below), locating the  $0.9182 \rightarrow 0.9635$  gap in quantisation flicker rather than model failure. Machine-generated by `reproduce.py --fig3`.

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